

The biologists' payoff is that they get to learn how cockroaches function. For instance, Joseph Ayers, a neuro-physiologist at Northeastern University in Boston, is now working on a robot lobster. He says, "The experiments you do on the robots tell you what you ought to be looking for within the animal." Some labs have reported that studying how insect bots fly led them to a conclusive explanation of how real insects fly—which, incidentally, is through complex and unconventional aerodynamics.

The Case Western experiment is a typical example of the intermeshing of disciplines that's happening in robotics. The program started with Randall Beer, a computer scientist, who was frustrated by the slow progress of AI. He sought guidance in the nervous systems of simple insects. However, animal bots are the tip of the iceberg when it comes to the New Wave. They may teach, but they don't learn.

Artificial life

It's only natural that a field called artificial life should have come up, since all scientists at some level want to play God. Artificial life comes from the idea that complex systems such as life, arise from simple interactions between enti-

ties in suitable environments. The field attempts to model as well as understand life using simulations that involve complex algorithms. For example, the Artificial Life Group at MIT discusses 'all things pertaining to the modelling and synthesis of biological systems.'

An ambitious artificial life project in 2002 gives a flavour of the kind of research that goes on: A team at the University of Zurich, Switzerland, created computer simulations with muscles, senses and nervous systems. These simulations were 'grown' from artificial embryos. The genome, or life code, for each of these was a string of random numbers. The creatures were real to the extent that their cells contained simulated, reactive chemicals. Josh Bongard, the head of the team, was hopeful that the creatures might develop brain-like regions when assigned tasks.

Appropriately, many of the algorithms used in such studies are genetic algorithms, which are a subset of what are known as 'evolutionary algorithms'. The concept of genetic algorithms comes from the evolution of life itself—survival of the fittest. People in artificial life labs often work with living tissue; but what about working with hardware? Hardware that knows itself, and can self-modify? Hardware that lives? That's

where evolvable hardware and evolutionary robotics come in—the marriage of robotics with artificial life.

Actually, there's a whole slew of technologies and disciplines that will make the smart bots of the future possible: traditional robotics, neural networks, evolutionary algorithms, programmable hardware, evolvable hardware, and many others. Every smart bot that's rolled out of a lab was born with the help of one or more of these techniques.

Robots that learn

Before the concepts of self-modifying machines, there was the idea that a robot could learn things—learn, adapt and improve. An example is Adam, a learning robot developed at Monash University, Australia. A niche environment called Eden was developed for Adam, with the expectation that he would learn and evolve there. Adam learnt to navigate within his environment; for example, how to avoid bumping into walls. That's already a major leap ahead from 'static' robots such as Aibo and Asimo, which learn only facts. As in artificial life, learning robots often use evolvable algorithms to drive their learning machinery, which, in hardware, is often a neural network of some sort (*See Box 'Training and Evolving Neural Networks', page 27*).

A bot-eat-bot world

If Adam doesn't sound interesting enough, here's something way out: in March 2002, at the Magna Science Adventure Centre in Rotherham, England, robots were divided into predators and prey, in a specially constructed environment. The aim of the experiment was to see if and how the community of bots would evolve an ecosystem. Wired magazine called it 'an experiment that sounds like a science-fiction film plot, but is actually as close to real life as artificial intelligence can get.' The people at Magna billed it as 'the largest experiment ever in artificial intelligence.'

It's not science fiction, but there's very little about the experiment—which ended in August 2003—to make it sound like science fact. The prey robots would have to feed by charging their solar batteries by sitting under artificial trees made of light. The predators would feed on the prey by hunting them down, immobilizing them, and insert-

Genetic Algorithms

Genetic algorithms are able to find optimal solutions to problems using the basic principles of Darwinian evolution—mutation, crossover, selection and so on. In many instances, they can find optimal solutions much faster than other algorithms.

Genetic algorithms work in the following manner: a chromosome in biology is a collection of genes. When working with genetic algorithms, think of a gene as a bit, and of a chromosome as a string of bits.

Each initial chromosome, in a genetic algorithm simulation, is taken to be a possible candidate solution. In this way, the 'solution', which is not actually a solution yet, is encoded as a string of bits. The chromosome, or string, may be made up of random genes—it can be a random string of bits. Now, with the initial collection of chromosomes at hand, each chromosome is tested for 'fitness', i.e., how good a solution it is. The ones that are better are more likely to reproduce.

Given the chromosomes that will reproduce, the crossover step comes in: two such chromosomes (remember that these chromosomes are strings of bits) are chosen, linked together, and at a particular point, are cut and joined each half to the other.

Mutation comes in at this point. After crossover, some bits (not too many) may be randomly changed, in order to introduce some variety. This is done once every few generations. The crossover and mutation steps are repeated as many times as required to produce the same number of chromosomes that were there in the initial population.

The new collection of chromosomes is again tested for fitness, just as with the initial population, and the cycle continues. So far, one 'generation' of the algorithm has been completed. The entire process is repeated until the collection of chromosomes obtained is a 'good enough' solution to the original problem.